

Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	VRF SYSTEM				
	MITSUBISHI ELECTRIC THAILAND				
	Supply, Installation, Testing and Commissioning of variable refrigerant flow modular type indoor units wired/wireless remote suitable for R410a refrigerant comprising of all accessories as per the specifications. The indoor units shall be suitable to work on cooling as well as heating mode. The minimum Energy Efficiency Ratio shall be as per ASHRAE STANDARDS 90.1. The indoor units shall be suitable for operation on 220 ±6% volts, 50Hz, 1 phase AC power supply. Ductable indoor units shall be suitable to handle extent of ductwork as shown in the design drawings and dehumidified air quantity as mentioned in the equipment schedule. The indoor unit shall be complete with pre filter & remote as per requirement The indoor units shall be of following capacities:				
	INDOOR UNITS				
	HIWALL UNITS				
A	B1-0B-IDU-01 of air flow rate 261 L/s and total cooling coil load of 5.7 kW	1	No		
B	B1-0B-IDU-02 of air flow rate 92 L/s and total cooling coil load of 1.2 kW	1	No		
C	B1-0B-IDU-03 of air flow rate 132 L/s and total cooling coil load of 1.7 kW	1	No		
D	B1-0B-IDU-05 of air flow rate 87 L/s and total cooling coil load of 1.1 kW	1	No		
E	B1-0B-IDU-06 of air flow rate 455 L/s and total cooling coil load of 8.7 kW	1	No		
F	B1-0B-IDU-07 of air flow rate 55 L/s and total cooling coil load of 2.8 kW	1	No		
G	B1-0B-IDU-08 of air flow rate 113 L/s and total cooling coil load of 1.4 kW	1	No		
H	B1-0B-IDU-09 of air flow rate 100 L/s and total cooling coil load of 1.3 kW	1	No		
J	B1-0B-IDU-11 of air flow rate 252 L/s and total cooling coil load of 7.2 kW		No		
			No		
			To Collection	QR.	-



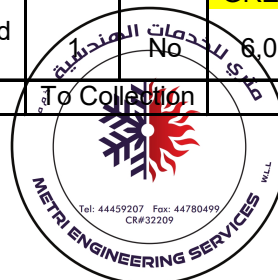
مركز الخدمات الهندسية

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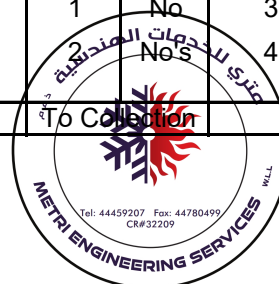
Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>REFRIGERANT PIPE INSTALLATION (Cont'd...)</u>				
A	Ø 12.7 mm	70	m	-	Included in Units
B	Ø 15.90 mm	45	m	-	Included in Units
C	Ø 19.05 mm	72	m	-	Included in Units
D	Ø 41.3 mm	35	m	-	Included in Units
	<u>DRAIN PIPING</u>				
	<u>Supply and installation of PVC-U drain pipe insulated with 19mm fibre glass pipe insulation.</u>				
E	Ø 20mm	138	m		
	<u>EXHAUST FANS</u>				
	<u>Supply Installation, testing & Commissioning of Low noise application exhaust fans, complete with motor, statically or dynamically balanced die-cast-in-aluminium impeller; rolled steel channel supporting frames, anti-vibration mountings, acoustic lining enclosure etc., complete as specified and indicated on the drawings, equipment schedule and technical specification of following capacities;</u>				
	<u>KITCHEN EXHAUST FAN</u>			GREENHECK	
	<u>Fan shall be listed in accordance with UL 762 or equivalent and suitable for commercial cooking</u>				
F	Exhaust Air Fan EF-B1-RF-02 of 807 L/s, Fire rated 400°C for 2 Hours	1	No	12,747.00	12,747.00
	<u>Supply Installation, testing & Commissioning of ELECTRO STATIC PRECIPITATOR as noted below</u>				-
	<u>Constant flow ESP unit with pre-filter, electrostatic precipitator and carbon filter for kitchen hood exhaust, Floor mounted type unit, etc. complete as specified and indicated on the drawings, equipment schedule, and technical specification of following capacities, Airflow 807 L/s</u>			FREGO	
G		1	No	19,500.00	19,500.00
	<u>CAR PARK EXHAUST FAN</u>			GREENHECK	-
H	Exhaust Air Fan EF-B1-0B of 1292 L/s, Fire rated 250°C for 2 Hours		No	6,080.00	6,080.00
	To Collection			QR.	38,327.00



Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	FIRE DAMPER <u>Supply, installation, testing and commissioning of Fire damper as per technical specification and local standards and as shown in the drawings including all accessories; fix in position with ductwork of following sizes.</u>				
A	250x100	2	No's	347.00	694.00
B	200x100	2	No's	330.00	660.00
C	150x150	8	No's	277.00	2,216.00
D	150x100	1	No	277.00	277.00
E	200x150	2	No's	287.00	574.00
F	250x150	1	No	297.00	297.00
G	200x200	1	No	294.00	294.00
H	450x500	1	No	416.00	416.00
J	600x500	1	No	455.00	455.00
K	300x150	1	No	306.00	306.00
L	300x200	1	No	315.00	315.00
M	350x200	2	No's	325.00	650.00
N	450x350	1	No	376.00	376.00
P	400x300	1	No	356.00	356.00
Q	700x400	1	No	446.00	446.00
R	600x450	3	No's	444.00	1,332.00
S	550x400	2	No's	410.00	820.00
T	550x450	3	No's	432.00	1,296.00
U	900x700	2	No's	480.00	960.00
V	900x650	2	No's	571.00	1,142.00
W	600x300	1	No	400.00	400.00
			To Collection	QR.	14,282.00



Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>VOLUME CONTROL DAMPER (CONT'D...)</u>				
A	450x350	2	No	235.00	470.00
B	500x200	2	No's	202.00	404.00
C	500x250	3	No's	214.00	642.00
D	500x300	1	No	219.00	219.00
E	500x350	3	No	243.00	729.00
F	500x400	2	No's	246.00	492.00
G	550x250	1	No	221.00	221.00
H	550x300	2	No	226.00	452.00
J	550x400	3	No	256.00	768.00
K	550x450	2	No	260.00	520.00
L	600x250	4	No	217.00	868.00
M	600x300	4	No's	222.00	888.00
N	600x450	2	No	253.00	506.00
P	650x400	1	No	288.00	288.00
Q	650x450	1	No	294.00	294.00
R	700x350	1	No's	294.00	294.00
S	750x350	3	No	305.00	915.00
T	800x300	2	No	276.00	552.00
U	800x500	2	No's	369.00	738.00
V	900x150	3	No	242.00	726.00
W	900x650	4	No's	453.00	1,812.00
X	900x700	2	No	456.00	912.00
Y	1000x300	1	No	308.00	308.00
Z	1000x650	2	No's	482.00	964.00
		To Collection		QR.	14,982.00



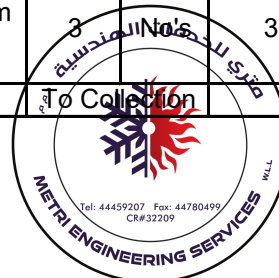
Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>BACK DRAFT DAMPER</u>				
	<u>Supply, installation, testing and commissioning of Back Draft Damper as per technical specification and local standards and as shown in the drawings including all accessories; fix in position with ductwork of following sizes.</u>				
A	200x150	2	No's	90.00	180.00
B	250x200	1	No's	122.00	122.00
C	300x200	2	No's	136.00	272.00
D	340x340	4	No's	202.00	808.00
E	440x440	4	No	237.00	948.00
F	700x500	1	No	207.00	207.00
G	1650x300	1	No	411.00	411.00
	<u>NON RETURN DAMPER</u>				
	<u>Supply, installation, testing and commissioning of Non Return Damper as per technical specification and local standards and as shown in the drawings including all accessories; fix in position with ductwork of following sizes.</u>				
H	150x100	1	No's	136.00	136.00
	<u>FRESH AIR SANDTRAP LOUVERS</u>				
	<u>Supply, installation, testing and commissioning of Fresh air inlets with Pre filter, 2" washable aluminium filter, High efficiency filter with powder coated finish to approved color, including sand trap louvers and insect screen / bird guard, sand trap as per specification.</u>				
J	500x400	1	No	500.00	500.00
K	600x450	2	No	500.00	1,000.00
L	600x500	2	No	500.00	1,000.00
M	1250x600	1	No	1,000.00	1,000.00
N	1650x450	1	No's	1,100.00	1,100.00
			To Collection	QR.	7,684.00



Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>EXTRACT AIR LOUVERS</u> <u>Supply, installation, testing and commissioning of Exhaust air louver with powder coated finish to approved color, including insect screen / bird guard, louver as per specification.</u>				
A	700x500	1	No's	397.00	397.00
B	1650x300	1	No	636.00	636.00
	<u>SOUND ATTENUATOR</u> <u>Supply, installation, testing and commissioning of Sound attenuators, including all accessories complete as per technical specification, local standards and as indicated on the drawings; jointing to ductwork (minimum length 1200mm) of following sizes in mm.</u>				
C	550 x 250	1	No	1,883.00	1,883.00
	<u>CROSS TALK ATTENUATOR</u> <u>Supply, installation, testing and commissioning of Cross talk attenuators, including all accessories complete as per technical specification and local standards. The size of the cross talk attenuator shall be followed as per the drawings.</u>				
D	800 x 300	1	No	900.00	900.00
	<u>MOTORIZED DAMPER</u> <u>Supply, installation, testing and commissioning of louver type motorised dampers confirm to SMACNA/NFPA in duct work with access door, actuator, matching flanges, gaskets & bolts, etc.. The item shall be complete including all the fittings and accessories as detailed in drawings and technical specifications and local standards. The actuator shall be ON/FF type. The actuator shall be capable of accepting 240 V AC supply, the controller shall be BMS ready and communicable to BACNET protocol with communication port. The item should be complete in all respect including all fittings, supports and accessories required for its complete installation and proper functioning as per standards, equipment schedule, technical specifications and details in drawings of following sizes.</u>				
E	500x400	1	No	1,500.00	1,500.00
		To Collection		QR.	5,316.00



Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>MOTORIZED DAMPER (CONT'D...)</u>				
A	600x300	1	No	1,500.00	1,500.00
B	600x450	2	No	2,500.00	5,000.00
C	600x500	2	No	2,640.00	5,280.00
	<u>FLEXIBLE DUCT</u>				-
	<u>Supply, installation and testing of Flexible duct not exceeding 1.5m, as per technical specification and local standard.</u>				-
D	Ø 200mm	48	m	50.00	2,395.00
E	Ø 150mm	78	m	40.00	3,120.00
	<u>BELL MOUTH WITH WIRE MESH</u>				-
	<u>Supply, fabrication, and installation of bell mouth with wire mesh, complete with necessary flanges, supports, and fixing accessories. The wire mesh shall be of aluminium with a minimum mesh size of [8-13] mm. All components to be as per project specifications and Tender drawings. The size of the bell mouth with wire mesh shall be followed as per the drawings.</u>				-
F	<u>AIR INLETS AND OUTLETS</u>				
	<u>Supply, installation, balancing/testing of Supply, return and exhaust air grilles and diffusers along with insulated plenum boxes (as applicable) as specified in the technical specification, local codes and standards and drawings, extruded aluminium construction with powder coated finish to approved colour; complete with supports and flexible connections; fixing in position including connections to ductwork as per standards, equipment schedule, technical specifications and details in drawings of following sizes.</u>				
	<u>Supply air linear diffuser with dampers, sizes as follows</u>				
G	2Slot Linear Diffuser, 1000mm length and 20 mm wide	16	No's	300.00	4,800.00
H	3Slot Linear Diffuser, 1000mm length and 25 mm wide	3	No's	300.00	900.00
				QR.	24,245.00



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Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>SUNDRIES IN CONNECTION WITH</u> <u>AIRCONDITIONING AND VENTILATION</u> <u>INSTALLATION.</u> <u>Allow for the following.</u>				
A	Marking the position of holes, mortices, chases and the like in the structure.	1	Item	EXCLUDED	EXCLUDED
B	Plates, discs, labels, tagging and charts for identification of plant, equipment and the like.	1	Item	EXCLUDED	EXCLUDED
C	Loose keys, tools and spare parts as specified.	1	Item	EXCLUDED	EXCLUDED
D	Temporarily operating, the Installations and assisting the CLIENT in the operations and maintenance of the Installations.	1	Item	INCL	INCL
E	Training to Client Personnel on HVAC System	1	Item	INCL	INCL



Item	Description	Qty	Unit	Rate (QR)	Amount (QR)
	<u>Collection:</u>				
	Page 2/8/1				260,000.00
	Page 2/8/2				-
	Page 2/8/3				171,550.00
	Page 2/8/4				99,833.00
	Page 2/8/5				38,327.00
	Page 2/8/6				270,209.50
	Page 2/8/7				142,090.00
	Page 2/8/8				14,282.00
	Page 2/8/9				52,439.00
	Page 2/8/10				14,982.00
	Page 2/8/11				7,684.00
	Page 2/8/12				5,316.00
	Page 2/8/13				24,245.00
	Page 2/8/14				65,079.00
	Page 2/8/15				69,000.00
	Page 2/8/16				-
	Page 2/8/17				-
	<u>TOTAL FOR HVAC INSTALLATIONS</u>			QR.	1,235,036.50

